Subclinical mastitis in water buffaloes in Bangladesh: prevalence and risk factors

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Bangladesh

Buffaloes in Bangladesh

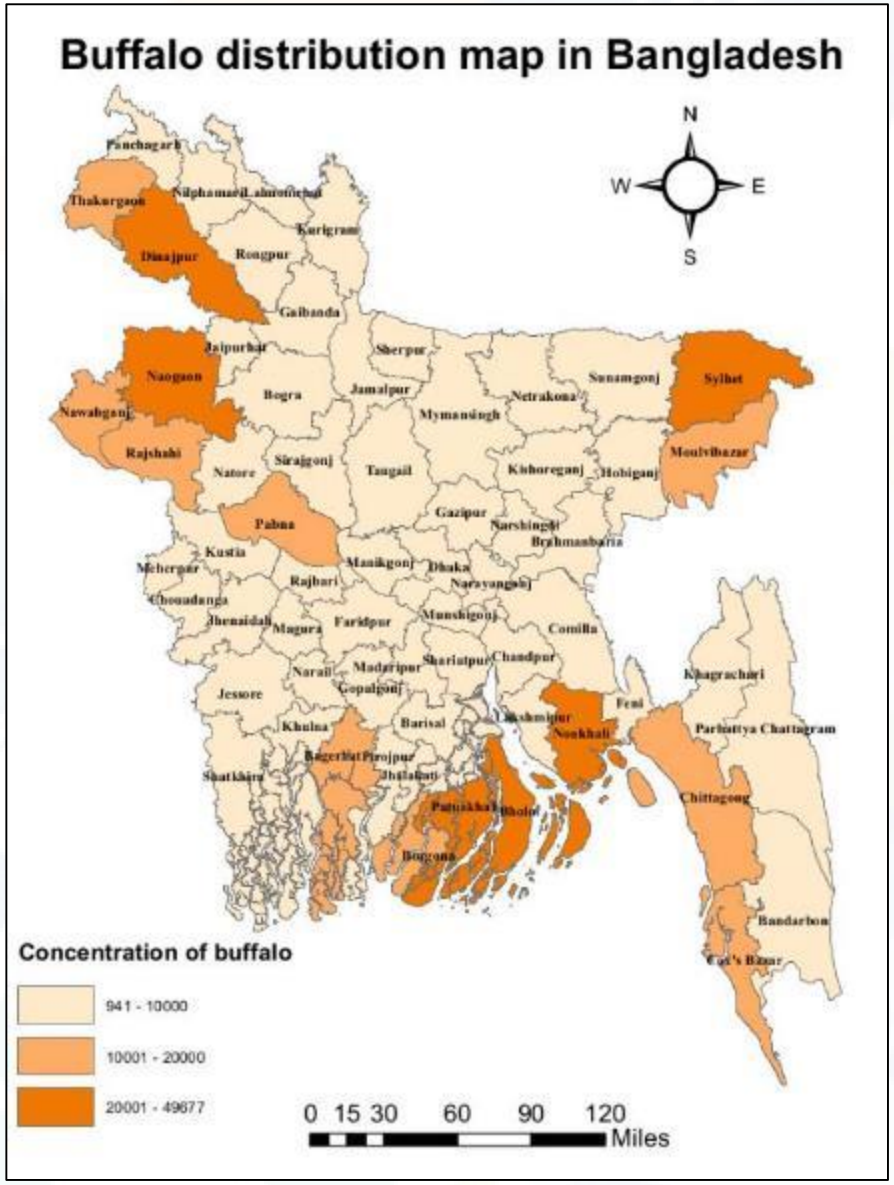
- 1.5 million buffaloes (2018-19) (DLS, 2020)
- Concentrated in the coastal areas
- Mainly indigenous breeds and poor milk producers
- Scattered cross breeds are available in few intensive farms in some areas

Indigenous

Murrah

Jafrabadi

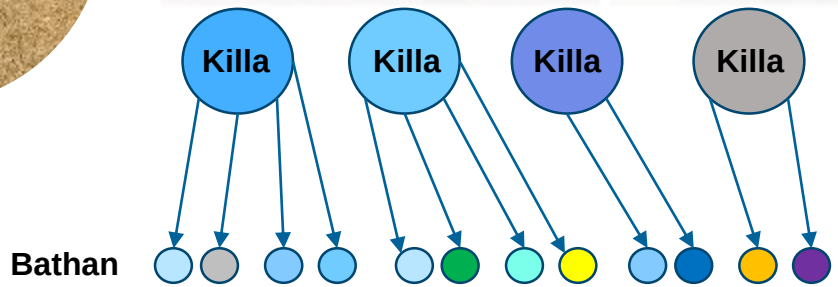
Nili-Ravi



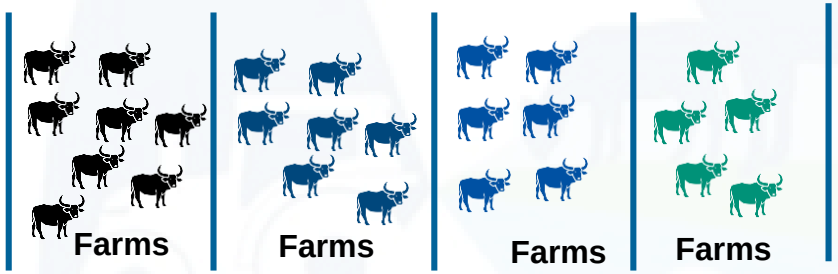
Map Prepared by Dr. Sanjib and Dr. Salma, CVASU & UHB, Bangladesh

Buffalo rearing in Bangladesh

Island



Inland



In Bathans

- Each farmer have 20-1000 buffaloes, cows and sheep in a “bathan”
- Buffaloes graze whole day and not provided any concentrate supplements
- There is no shed for the buffaloes, they stay in open island

In Inlands

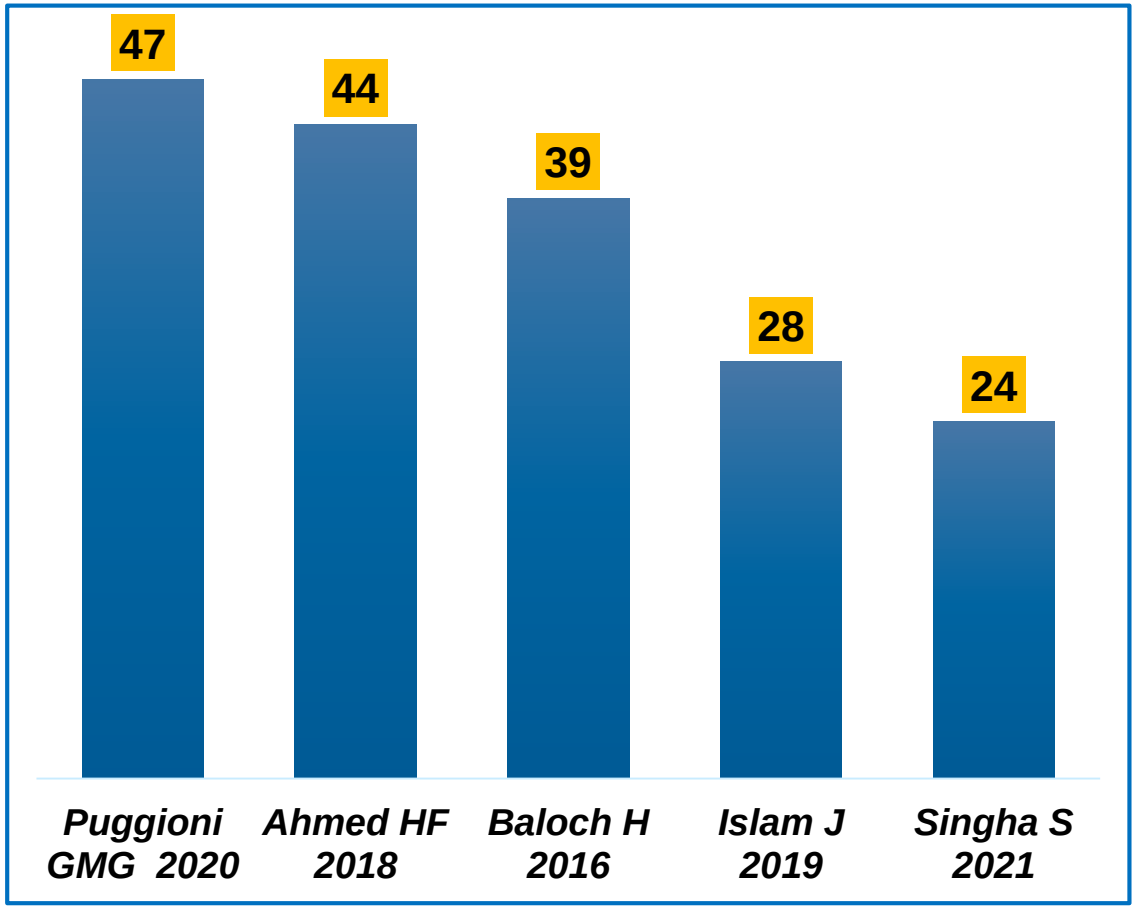
- **Semi-bathan:** Buffaloes come back from bathan and stay in inlands for certain period (3-6 months)
- **Semi-intensive:** Buffaloes stay at sheds, go to graze and come back everyday
- **Household:** Buffaloes stay at shed or under tree and allowed to graze
- **Intensive:** Buffaloes tied at sheds, stall fed and are never allowed to graze

Subclinical mastitis in buffaloes

Mastitis is a major challenge for dairy buffaloes

- **Animal comfort is ignored [1]**
- **Fluctuation in surrounding temperature and humidity [2]**
- **Insufficient milking hygiene [1]**
- **Feed scarcity [1]**

Buffalo SCM prevalence



1. *Alam, M. R., Islam, M. J., Amin, A., Shaikat, A. H., Pasha, M. R., & Doyle, R. E. (2020). Animal-Based Welfare Assessment of Cattle and Water Buffalo in Bangladeshi Slaughterhouses. Journal of Applied Animal Welfare Science, 23(2), 219-230.*
2. *Ahmad, Sibtain & Tariq, Muhammad. (2010). Heat stress management in water buffaloes: A Review. Revista Veterinaria. 21. 301-314.*

Objectives of the study

- Estimate the prevalence of subclinical mastitis (SCM) in buffaloes
- Estimation of bulk milk somatic cell count (BMSCC) levels
- Identification of buffalo level risk factors for SCM
- Identification of quarter specific risk factors for SCM
- Identification of intra-mammary infection (IMI) causing pathogens

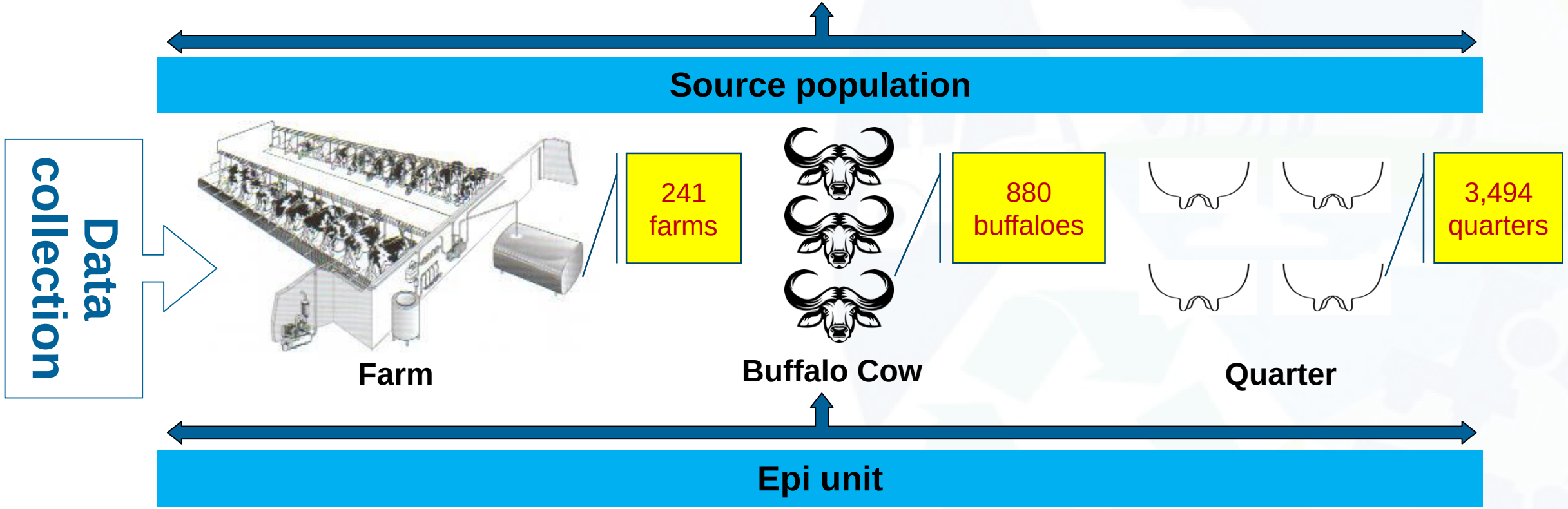
Sampling

- ❑ **Lactating buffaloes**
- ❑ **Functional quarters**
- ❑ Selected buffalo pockets in Bangladesh

Study period

September-October 2019

February, 2020-March, 2021



Detection of subclinical mastitis in buffaloes using California mastitis test (CMT)

CMT of milk from each quarter

↓

Scoring for milk for SCM detection
(Score 1-5)

↓

Aseptically collection and storage of quarter and bulk milk samples

↓

Bulk milk somatic cell count (BMSCC)

↓

Transportation of samples to CVASU lab and bacterial culture of quarter samples for identification of IMI pathogens

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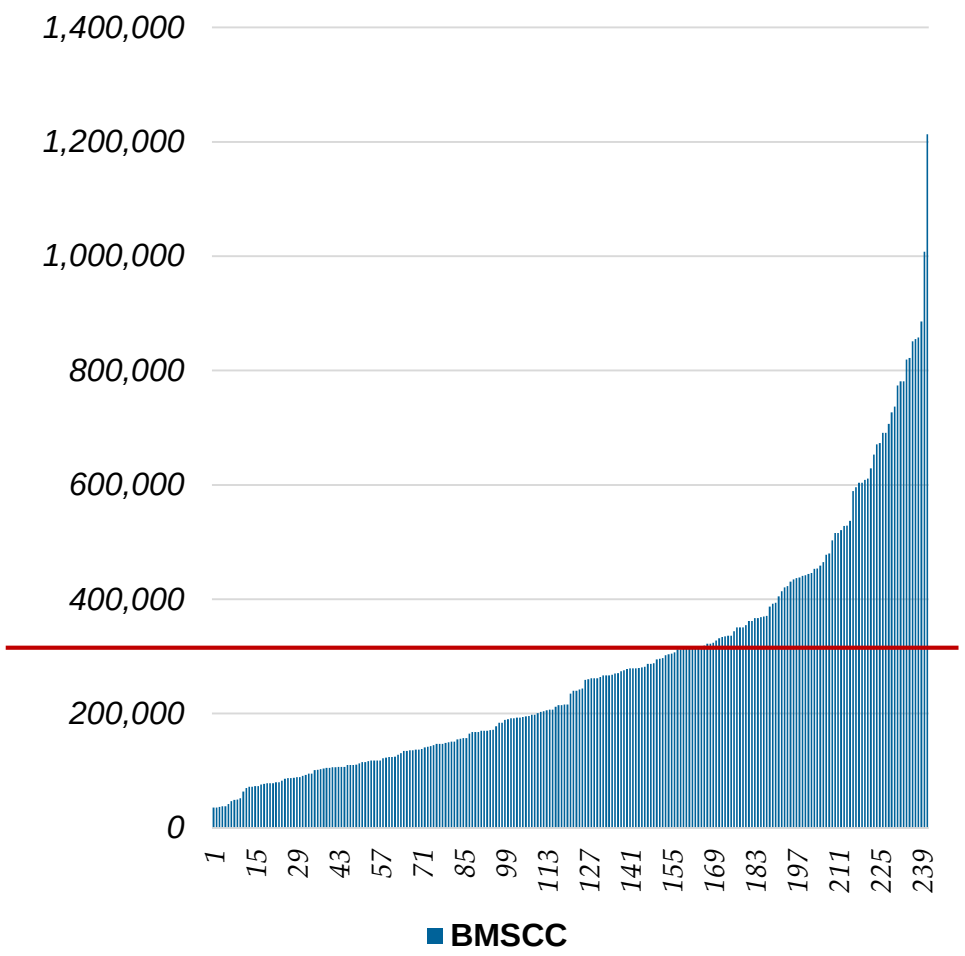
Isolation and identification of pathogens using bacteriological culture

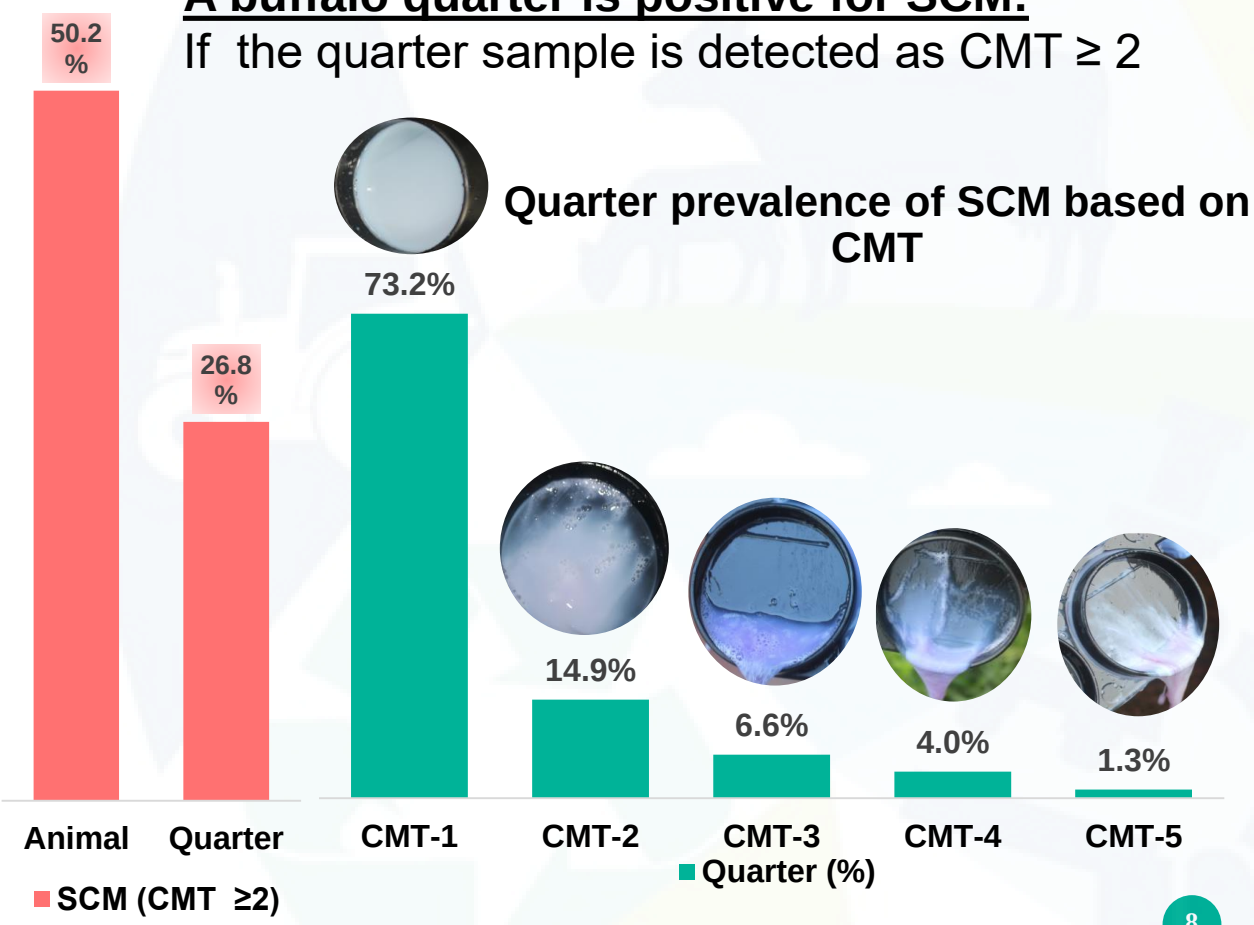
Prevalence of SCM in buffaloes



Mean BMSCC: 282,187/mL of milk

A buffalo is positive for SCM:
If at least one quarter of the animal gets CMT ≥ 2

A buffalo quarter is positive for SCM:
If the quarter sample is detected as CMT ≥ 2



Animal level risk factors

Multivariable logistic regression model using farm as a random effect

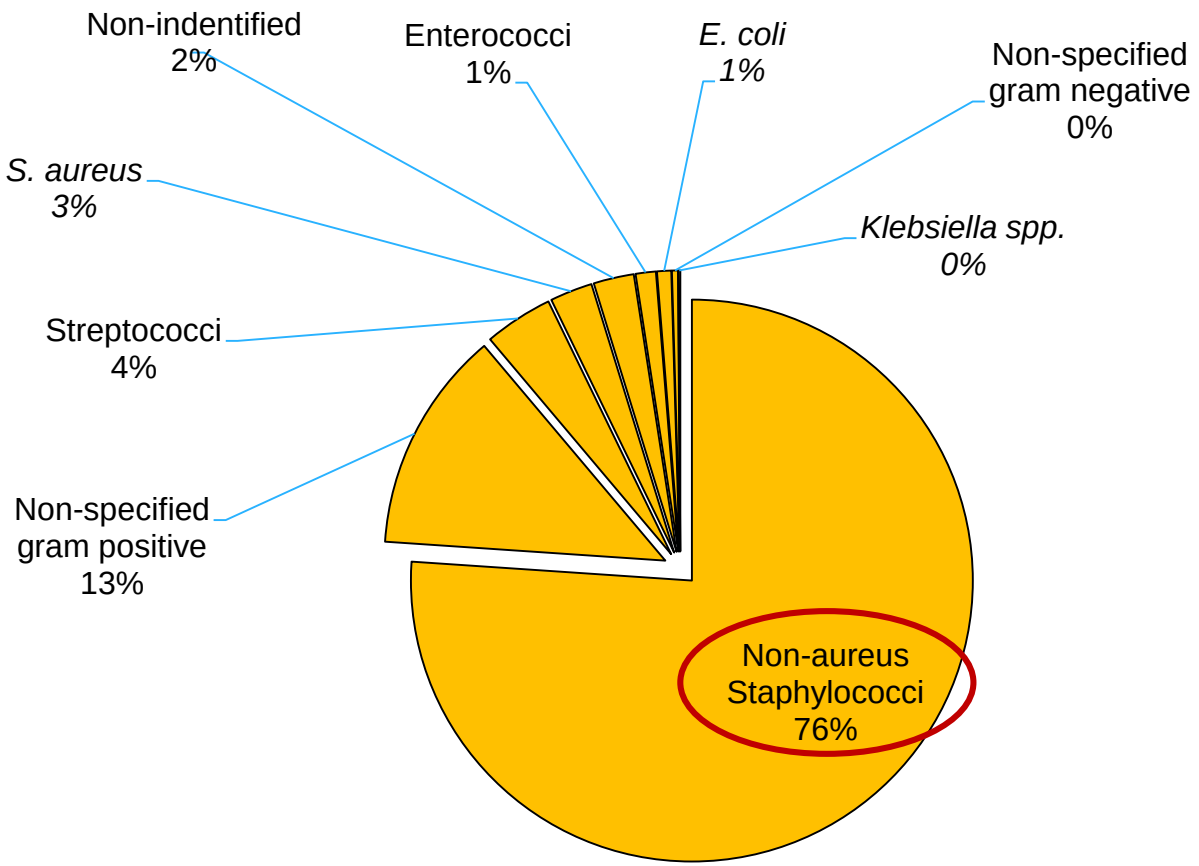
Variable name	Categories	Odds Ratio (95% CI)	P
Farm type	Semi-intensive	1.0	1.0
	Semi-bathan	1.2	0.2
	Bathan	Ref	
	Household	1.2	0.6
	Intensive	2.6	0.02
Age (years)	2.5 to 6	1.4	0.1
	6.1 to 8	Ref	
	8.1 to 20	1.3	0.2
Teat position	Symmetrical	Ref	
	Asymmetrical	2.0	<0.001
History of calf mortality	Yes	1.9	0.06
	No	Ref	
Person do the milking	Single	1.5	0.008
	Multiple	Ref	
Source of buffaloes	Own stock	1.8	0.08
	Purchased	1.8	0.1
	Take care of others buffaloes	Ref	

Quarter level risk factors

Multivariable logistic regression model using farm and animal as a random effect

Variable name	Categories	Odds Ratio (95% CI)	P
Farm type	Semi-intensive	1.1 (0.6 to 2.2)	0.8
	Semi-bathan	1.1 (0.6 to 2.0)	0.8
	Bathan	Ref	
	Household	2.1 (0.9 to 4.9)	0.1
	Intensive	7.0 (2.1 to 23.1)	0.001
Quarter position	Left quarters	1.4 (1.1 to 1.7)	0.004
	Right quarters	Ref	
Teat shape	Cylindrical	Ref	
	Bottle	1.3 (0.9 to 2.0)	0.2
	Funnel	2.1 (1.4 to 3.3)	0.001

Pathogen (N=863) distribution in all buffalo milk samples collected from study area



Conclusion

1. High prevalence of animal and quarter level subclinical mastitis

2. Mean bulk milk somatic cell count is quite satisfactory is an indication of good buffalo milk quality and possibilities to adopt management following better farms

3. SCM was most common in intensive farms so careful assessment of buffalo management is needed in this rearing system in Bangladesh and also applicable to the similar management system in other countries

4. For young or older animals with asymmetrical teats, restricted hygienic measures are necessary

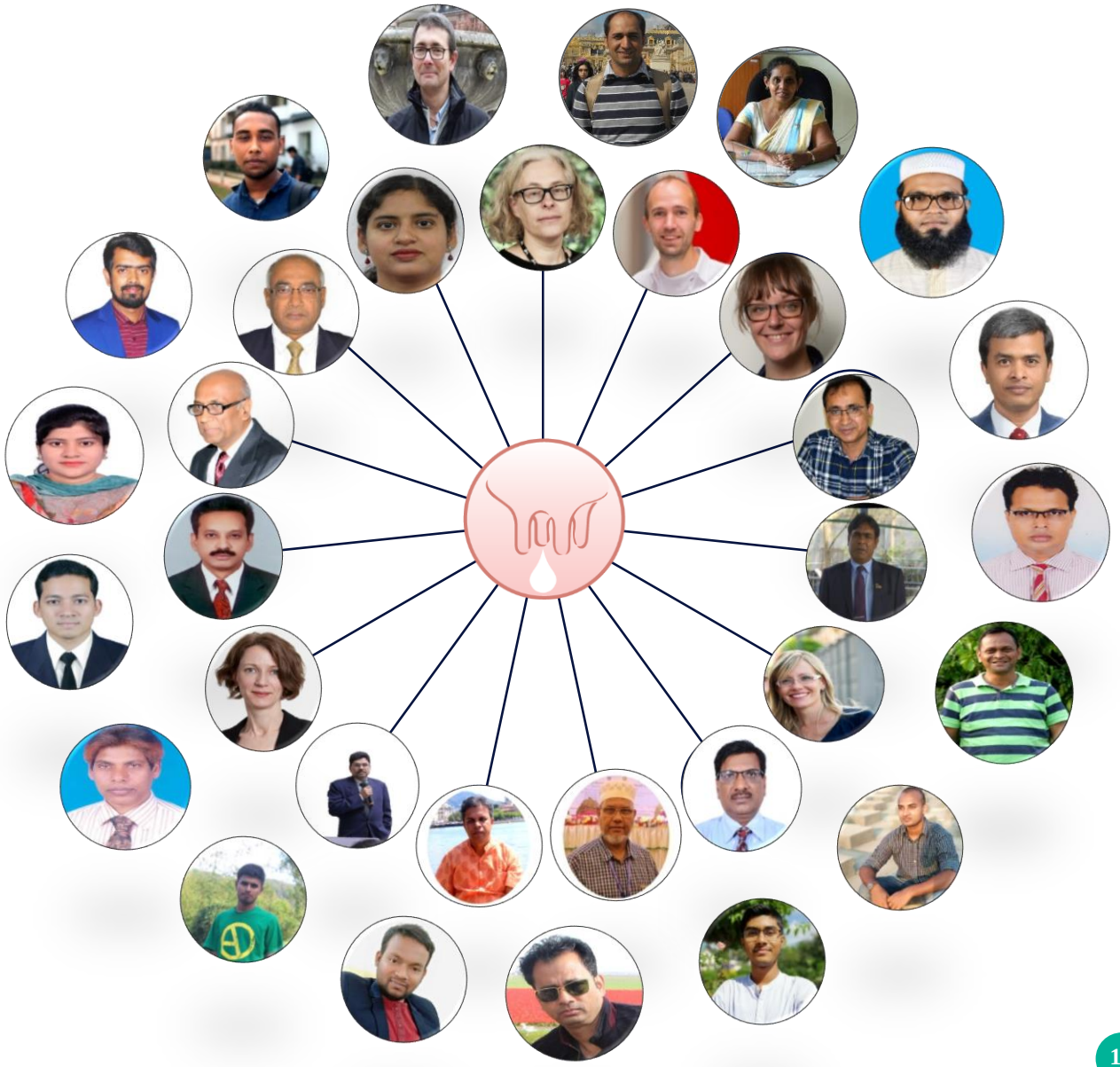
5. Left quarters, bottle and funnel shaped quarters, hygienic milking practices should be strictly followed to prevent IMI thereby reduce SCM

6. Non-aureus Staphylococci appeared as the most frequent IMI causing pathogens which need to be further identified at species level along with virulence characteristics

Further studies to be conducted

- Identification of the bacteria at species level through MALDI-TOF
- Antimicrobial resistance of the *Staphylococcus* spp.
- Assessment of the genetic diversity and virulence characteristics of *Staphylococcus* spp.

Udder health Bangladesh Team



Thank you

***E. coli* in Violet Red Bile Glucose Agar**